

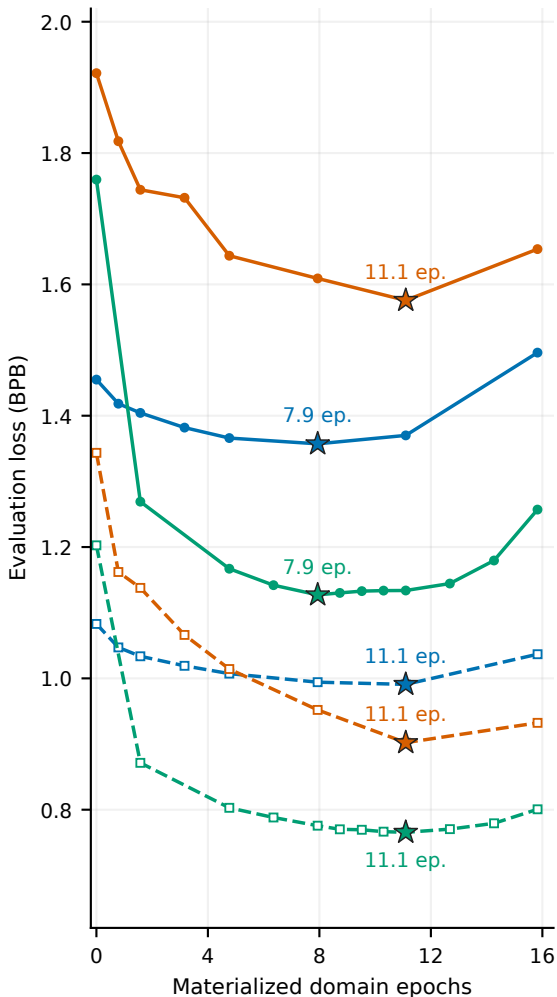
Wikipedia FineMath-3+ StarCoder

Simulated-epoching proxy (2.49×10^{16} FLOPs) Target (6.66×10^{18} FLOPs)

★ Measured minimum ♦ Fixed epoch count (1.6, 4.8, 7.9, 11.1) ◆ Simulated-epoching selection

Proxy and target sweeps

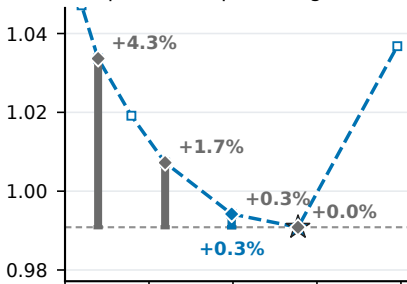
Wikipedia: Wikipedia English | FineMath-3+: GSM8K
StarCoder: Paloma code



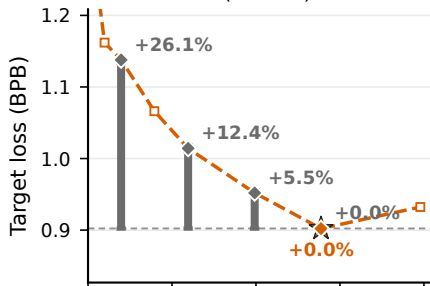
Fixed epoch count vs. simulated epoching

Excess target loss over measured optimum

Wikipedia (Wikipedia English)



FineMath-3+ (GSM8K)



StarCoder (Paloma code)

